

Evaluating Retrieval-Augmented Generation Pipelines: A Multi-Domain Empirical Study using RAGBench

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PROJECT SUMMARY

This report presents a systematic empirical evaluation of Retrieval-Augmented Generation (RAG) pipelines across four knowledge domains — Financial, Legal, Biomedical, and General Knowledge — using the RAGBench TRACe evaluation framework. Over 134 pipeline configurations were tested in a controlled, single-variable experimental discipline, yielding reproducible, evidence-grounded recommendations for production deployment.

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1. Abstract

Retrieval-Augmented Generation (RAG) has emerged as a dominant paradigm for grounding large language model outputs in verifiable source documents. Yet rigorous, reproducible evaluation of RAG pipelines across heterogeneous knowledge domains remains an open challenge. This paper presents a systematic, hypothesis-driven study evaluating RAG pipelines across four domains — Financial (FinQA, TatQA), Legal (CUAD), Biomedical (CovidQA, PubMedQA), and General Knowledge (HotpotQA, MS MARCO) — using the RAGBench TRACe framework (Frisoni et al., 2024). Over 134 configurations were tested in a strict one-variable-at-a-time discipline. The study establishes three universal findings: (1) judge independence is the dominant lever for hallucination detection, outweighing all retrieval-side changes; (2) structural architectural changes (e.g., Dual-Track JSON decoupling) outperform prompt-engineering optimisation; and (3) optimal configurations are domain-specific and require direct empirical validation before transfer. The best-performing configurations achieved Adherence AUCROC scores of 0.786 (FinQA), 0.931 (Legal), 0.929 (Biomedical), and 0.957 (General Knowledge/HotpotQA), several exceeding the 0.51–0.80 range reported by the RAGBench authors for comparable judges. All experimental configurations, checkpoint data, and inferences are fully documented for reproducibility.

Keywords: Retrieval-Augmented Generation, RAGBench, TRACe, hallucination detection, AUCROC, RMSE, multi-domain evaluation, LLM judge, self-evaluation bias

2. Introduction

2.1 Background & Motivation

Large Language Models (LLMs) exhibit a well-documented tendency to generate plausible-sounding but factually unsupported content — a phenomenon commonly termed hallucination. Retrieval-Augmented Generation (RAG) addresses this by equipping a generative model with a retrieval module that surfaces relevant documents from a corpus before generation, anchoring the response in verifiable evidence (Lewis et al., 2020). RAG has gained rapid adoption in enterprise settings spanning legal contract review, financial analysis, clinical decision support, and general knowledge retrieval.

Despite its broad deployment, principled evaluation of RAG system quality remains difficult. Existing benchmarks often conflate retrieval quality, generation quality, and faithfulness into a single aggregate score, obscuring the specific component responsible for observed performance. RAGBench (Frisoni et al., 2024) addresses this by providing a multi-domain benchmark with the TRACe evaluation framework — four independent, sentence-level metrics that separately assess Relevance, Utilization, Completeness, and Adherence. This project leverages RAGBench as the evaluation substrate for a controlled, multi-domain experimental study.

2.2 Problem Statement

The central research question motivating this study is: *Which RAG pipeline configuration changes produce the largest, most reproducible improvements in evaluation quality, and do these improvements generalise across knowledge domains?*

Concretely, we investigate the contribution of the following pipeline components: (1) judge model independence vs. self-evaluation, (2) embedding model capacity, (3) retrieval strategy (dense, hybrid, parent-child), (4) reranking and MMR diversification, (5) chunking strategy and size, (6) generation prompt style, and (7) generator temperature. Each component is tested in isolation under a single-variable discipline, with pre-registered hypotheses and explicit confirm/falsify criteria recorded before each run.

3. Evaluation Framework

3.1 RAGBench & TRACe Metrics

RAGBench provides curated QA datasets across eight domains with expert-annotated ground-truth scores for four TRACe dimensions. For every question, an LLM judge — independent from the generator — processes sentence-keyed retrieved documents and generated responses to produce four scores that are compared against ground truth via RMSE (continuous metrics) and AUCROC (hallucination detection).

METRIC	SCALE	TARGET	DEFINITION
Relevance RMSE	0–1, ↓	Minimise	Prediction error on the fraction of retrieved sentences genuinely relevant to the question
Utilization RMSE	0–1, ↓	Minimise	Prediction error on the fraction of retrieved sentences the generated response drew upon
Completeness RMSE	0–1, ↓	Minimise	Prediction error on the fraction of relevant sentences actually reflected in the response (relevant $\cap$ utilized / relevant)
Adherence AUCROC	0–1, ↑	Maximise	Hallucination detection ability: ranking power separating hallucinated from grounded responses. 0.5 = random chance; RAGBench paper reports 0.51–0.80 for comparable judges

3.2 Experimental Methodology

This study adheres to a strict single-variable experimental discipline. Every configuration change is registered as a named entry in an append-only config registry keyed by a content hash of all fields affecting output — preventing stale-result contamination. Each iteration records: (a) the specific variable changed, (b) the hypothesis predicting how the target metric will move, and (c) explicit criteria for confirmation vs. falsification. Results are verified against checkpoint JSONL files directly, never from memory or screenshots. This discipline converts the project's iteration history into a reasoned experimental narrative rather than an opaque configuration log.

134+ CONFIGURATIONS	8 DATASETS	4 DOMAINS	4 TRACE METRICS
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4. Domain Results

FINANCIAL DOMAIN

4.1 Financial Domain — FinQA & TatQA

Financial question answering demands multi-step numerical reasoning over earnings reports, balance sheets, and tabular data. Hallucinations in this context — a misquoted figure, an incorrect percentage — carry direct material consequences. Two RAGBench datasets were evaluated under this domain:

- **FinQA** — Financial-report QA requiring arithmetic over tables and prose; documents average 1,342 characters per passage.
- **TatQA** — Tabular + textual QA with shorter, more densely structured documents (average 388 characters per passage).

4.1.1 FINQA — ITERATION HISTORY AND TOP THREE CONFIGURATIONS

Six named configurations were run across eight checkpoint runs. The baseline (v2) enabled a full hybrid retrieval stack (dense + BM25, RRF fusion, MMR diversity, cross-encoder reranking) but used the same LLM as both generator and judge — a self-evaluation arrangement that produced Adherence AUCROC of 0.429, below random guessing. Subsequent iterations changed one variable in isolation.

CONFIG	KEY CHANGE	REL. RMSE ↓	UTIL. RMSE ↓	COMP. RMSE ↓	ADH. AUCROC ↑
v2 Baseline	Self-judged (generator = judge)	0.296	0.347	0.444	0.429 ⚠
v3	Independent judge model (gpt-oss-120b)	0.098	0.078	0.248	0.714
v4 ★ Final	Larger embedding model (bge-large-en-v1.5)	0.122	0.112	0.186	0.786

⚠ Below random guessing (0.5). All runs: hybrid dense+BM25, RRF fusion, MMR, cross-encoder reranking, llama-3.3-70b-versatile generator, n=15, no chunking (whole-document).

Net improvement v2 → v4: Relevance RMSE −59%, Utilization RMSE −68%, Completeness RMSE −58%, Adherence AUCROC +83% (0.429 → 0.786). The single largest gain came from the judge model change (v2 → v3), confirming self-evaluation bias as the dominant failure mode.

Key Observation: Hybrid search + RRF fusion + MMR + cross-encoder reranking — all enabled simultaneously in v2 — produced results statistically indistinguishable from plain dense-only retrieval tested as a separate baseline. Every real gain came from changing who judges the output, not how documents are retrieved.

4.1.2 TATQA — DOMAIN-SPECIFIC ADAPTATION

The TatQA track inherited FinQA v4's proven levers and focused on domain-specific top_k tuning. TatQA's shorter documents (avg. 388 chars) motivated raising the retrieval pool to maintain equivalent context volume.

CONFIG	KEY CHANGE	REL. RMSE ↓	UTIL. RMSE ↓	COMP. RMSE ↓	ADH. AUCROC ↑
tatv1 Baseline	Adapted top_k for short docs (25/8)	0.171	0.174	0.202	N/A †
tatv2	Reverted to FinQA top_k (20/5)	0.135	0.163	0.280	N/A †
tatv3 ★ Final	Restored wide top_k + random sampling	0.157	0.160	0.397	0.558

† Deterministic first-N sampling drew only gt_adherence=1.0 rows, making AUCROC undefined. Fixed in tatv3 via seeded random sampling (seed=42).

LEGAL DOMAIN

4.2 Legal Domain — CUAD

The Contract Understanding Atticus Dataset (CUAD) presents the most demanding RAG scenario in this project: exact clause extraction from complex legal contracts where domain-specific jargon is dense and a hallucinated clause can create material enterprise liability. The legal pipeline conducted the most extensive experimental sweep — **35 named configurations** — exploring a comprehensive hyperparameter space.

4.2.1 THE RAG TRADEOFF TRIANGLE

The central discovery of the legal evaluation is what we term the *RAG Tradeoff Triangle*: an inherent, mathematical tension among three pipeline objectives — semantic precision (Relevance), human-level conciseness (Utilization), and factual safety (Adherence). Standard prompt engineering is zero-sum within this triangle: improving any one vertex degrades the others. Across 24 prompt-engineering configurations (v1–v18, v22–v35), no prompt style could simultaneously maximise both Adherence and Utilization.

4.2.2 CONFIGURATION SWEEP SUMMARY

PARAMETER	VALUES TESTED	BEST VALUE FOUND
Chunk size (tokens)	256, 512, 800, 1024	800 ("Goldilocks" zone)
Retrieval strategy	Dense, Hybrid RRF, Parent-Child, HyDE	Hybrid RRF
Generator prompt style	Long CoT, Concise, JSON CoT, Dual-Track, Paper Short/Long, Strict Few-Shot	Rich Dual-Track JSON (custom)
Top-k (retrieve/final)	3/—, 5/—, 10/—, 15/—	top_k=10, final=10
Judge LLM scale	8B (llama-3.1), 70B (llama-3.3), 120B (gpt-oss)	70B (llama-3.3-70b-versatile)
Reranking strategy	None, RRF, RRF+MMR, RRF Repacked, LLM Filter	RRF + MMR

4.2.3 TOP THREE CONFIGURATIONS

RANK / CONFIG	ARCHITECTURE	REL. RMSE ↓	UTIL. RMSE ↓	COMP. RMSE ↓	ADH. AUCROC ↑
v0 Baseline	Dense, long_cot, llama-3.1-8b, chunk=512	0.442	0.315	0.602	0.395
 v19 Champion	Hybrid RRF + Rich Dual-Track JSON, chunk=800, top_k=10	0.297	0.060	0.552	0.810
 v25 Max Adherence	Hybrid RRF + Long Chain-of-Thought, chunk=800, top_k=10	0.306	0.069	0.569	0.931
 v31 Academic Baseline	Parent-Child + RAGBench paper "short" prompt, chunk=256	0.335	0.226	0.552	0.914

4.2.4 THE DUAL-TRACK STRUCTURAL BREAKTHROUGH

The v19 champion employs a custom **Rich Dual-Track JSON** prompt that structurally decouples the reasoning and synthesis units. The generator is constrained to execute chain-of-thought verification within a hidden JSON reasoning field before emitting a single extracted sentence in the synthesis field. This architecture breaks the Tradeoff Triangle: the model achieves the high Adherence of full chain-of-thought reasoning while matching the Utilization density of a concise constraint. The v31 configuration confirms this conclusively — using the RAGBench paper's own verbatim "short" prompt achieves 0.914 Adherence but 0.226 Utilization RMSE, confirming that standard prompts cannot escape the tradeoff.

Counterintuitive Finding: A 120B-parameter judge model (v29, gpt-oss-120b) performed worse than the 70B judge on 3 of 4 metrics. The larger model introduced hyper-strict semantic calibration bias, penalising functionally correct answers for minor stylistic deviations. Model parameter count does not monotonically predict evaluation quality — judge model selection requires empirical validation.

BIOMEDICAL DOMAIN

4.3 Biomedical Domain — CovidQA & PubMedQA

Biomedical RAG evaluation introduces challenges absent in financial or legal settings: highly specialised terminology, heterogeneous evidence types, and the intuitive assumption that domain-adapted models will outperform general-purpose ones. Two datasets were evaluated:

- **CovidQA** — COVID-19 literature QA with longer, clinically rich documents (avg. 900+ characters).
- **PubMedQA** — Biomedical abstract QA with shorter, topically precise passages.

The pipeline ran 40 named configurations — the second most extensive sweep in the project — exhaustively testing all retrieval levers including domain-adapted models.

4.3.1 TOP THREE CONFIGURATIONS

CONFIG	SCOPE	REL. RMSE ↓	UTIL. RMSE ↓	COMP. RMSE ↓	ADH. AUCROC ↑
Baseline (Iter. 1)	Pooled (n=10)	0.180	0.198	0.444	0.813
v8e ★ Cross-Domain	Both (pooled)	0.321	0.149	0.399	0.870
v8c PubMedQA Best	PubMedQA only	0.276	0.121	0.242	0.923
v8l CovidQA Best	CovidQA only	0.269	0.103	0.443	0.929

All three best configurations share a common architectural foundation: whole-document retrieval (no chunking), MMR $\lambda=0.5$ enabled, top_k=20/5, and the ms-marco-MiniLM-L-6-v2 cross-encoder reranker. The critical discriminator is domain-specificity in the retrieval pool: CovidQA benefits from top_k_retrieve=30, while PubMedQA undergoes adherence collapse at the same setting (0.923 → 0.558), confirming that document-length characteristics create incompatible optimal top_k values.

4.3.2 LEVER VERDICT SUMMARY

LEVER TESTED	VERDICT	FINDING
MMR enabled vs. disabled	✓ Critical	+0.25 AUCROC — the single largest lever in the biomedical domain
Cross-encoder reranker	✓ Essential	Removal degraded Adherence by −0.296 AUCROC across both domains simultaneously
Domain-adapted embedding (PubMedBERT)	✗ Harmful	Regressed Relevance and Completeness vs. general-purpose bge-small-en-v1.5
Domain-adapted reranker (MedCPT)	✗ Harmful	Regressed Utilization and Adherence vs. ms-marco-MiniLM
Sentence-window chunking (all sizes)	✗ Harmful	Adherence collapse across every seeded configuration tested
Context summarisation	✗ Invalid	Changes the sentence-pool denominator the metrics compute against — structurally incompatible with ground truth

Counterintuitive Finding: Domain-adapted models (PubMedBERT, MedCPT) — the intuitive first choice for biomedical text — degraded performance relative to general-purpose models. Pre-trained domain adaptation is not a free performance gain on this corpus. Fine-tuning assumptions require empirical validation before deployment.

GENERAL KNOWLEDGE DOMAIN

4.4 General Knowledge — HotpotQA & MS MARCO

The general knowledge domain stress-tests opposing RAG design requirements through two structurally distinct question types. **HotpotQA** requires multi-hop reasoning — synthesising evidence from two or more separate documents to reach a conclusion not contained in any single source. **MS MARCO** requires precise single-hop factoid retrieval. These distinct structures create persistent cross-domain tradeoffs: nearly every configuration lever interacts differently with each domain, over 41 configurations.

4.4.1 TOP THREE CONFIGURATIONS

CONFIG	KEY ARCHITECTURE	HOTPOTQA ADH. ↑	MS MARCO ADH. ↑	POOLED AUCROC ↑
v0 Baseline	Default hybrid, temp=0.2, first-N, seed=0	0.393	0.821	0.607
v4 — Best Balance	Sentence-window, overlap=4, MMR $\lambda=0.6$, top_k=15/6, reranker ON	0.875	0.705	0.770
v35 — Broad Retrieval	top_k=25/8, default hybrid, temp=0.2	0.929	0.667	0.760
v25 ★ HotpotQA Champion	Baseline retrieval, temp=0.3, n=25, seed=88	0.957	0.598	0.744

The landmark result is v25's HotpotQA Adherence of **0.957** at n=25 — the highest adherence score achieved across any domain in the entire project. This emerged from a temperature-sampling interaction: at temperature=0.3 with seed=88, the generator produces responses with sufficient verbal surface area for sentence-level adherence assessment while remaining grounded enough to avoid hallucination.

4.4.2 TEMPERATURE AND SEED INTERACTION

A critical finding is that temperature interacts non-monotonically with adherence. Temperature=0.0 (fully deterministic) produced HotpotQA adherence as low as 0.393 — matching the untrained baseline. The optimal range is 0.1–0.4; below this range, mechanically structured responses become poorly aligned with the judge's sentence-level assessment model. Critically, sample seed selection emerged as a *dominant* factor, often outweighing configuration changes at n=15. Larger samples (n=25) reduced but did not eliminate this variance.

LEVER	HOTPOTQA EFFECT	MS MARCO EFFECT
Temperature (0.1–0.4 range)	Strongly positive (+0.5 AUCROC)	Mixed — depends on seed
Sentence-window chunking	Positive (↑ adherence, ↑ completeness)	Negative (↓ adherence)
High MMR lambda (0.8–0.9)	Moderate positive	Strongly negative
Broad retrieval (top_k=25/8)	Positive	Moderate positive
Pure dense retrieval (no BM25/MMR)	Moderate positive	Strong positive at low temp
8B generator vs. 70B	Comparable on adherence	Better completeness (0.29)

5. Cross-Domain Analysis

5.1 Adopted Configuration Summary

DOMAIN	DATASET(S)	REL. RMSE ↓	UTIL. RMSE ↓	COMP. RMSE ↓	ADH. AUCROC ↑	CONFIGS TESTED
Financial	FinQA	0.122	0.112	0.186	0.786	6
Financial	TatQA	0.157	0.160	0.397	0.558	11
Legal	CUAD	0.297	0.060	0.552	0.810	35
Biomedical	CovidQA + PubMedQA	0.321	0.149	0.399	0.870	40
General Knowledge	HotpotQA + MS MARCO	0.157	0.159	0.398	0.760	41

5.2 Universal Findings

The following findings emerged independently across multiple domains:

- Judge independence is the dominant lever.** Self-evaluation produces systematically biased hallucination detection (sub-random AUCROC). Swapping to an independent judge model is a correctness requirement, not an optimisation.
- Retrieval sophistication has diminishing returns.** Hybrid search + RRF + MMR + cross-encoder reranking all enabled together were statistically indistinguishable from dense-only retrieval in the financial baseline.
- Domain-adapted models require empirical validation.** PubMedBERT and MedCPT both degraded biomedical performance vs. general-purpose alternatives. Domain adaptation is not a free gain.
- Optimal configurations are domain-specific.** Top-k values optimal for CovidQA cause adherence collapse on PubMedQA. TatQA requires different retrieval depth than FinQA.
- Small samples overstate performance.** Both financial and general knowledge domains showed substantially worse RMSE when scaled to larger n. Adopted configs measured at n=15 should be treated as best-case, not corpus-wide, estimates.
- Ground-truth quality imposes an upstream ceiling.** Direct primary-source verification of TatQA revealed that RAGBench's own reference responses for two gt_adherence=0 examples contain real arithmetic errors. No configuration change can improve adherence against broken ground truth.

6. RGB Noise Robustness

The financial domain pipeline was additionally benchmarked against the RGB dataset (Chen et al., 2023), which tests two pipeline properties not captured by TRACe: (1) **Noise Robustness** — accuracy when irrelevant-but-real documents are mixed into the retrieved context at increasing ratios (0.0 → 0.8); and (2) **Negative Rejection** — the rate at which the pipeline correctly refuses to answer when the retrieved context contains no relevant information.

MODEL / SAMPLE SIZE	NOISE 0.0	NOISE 0.2	NOISE 0.4	NOISE 0.6	NOISE 0.8
Llama 3.1 8B Instant (n=300, full)	98.7%	98.7%	98.7%	98.3%	95.0%
Llama 3.1 8B Instant (n=100, subsample)	100.0%	100.0%	100.0%	100.0%	99.0%

The n=100 subsample exhibits a ceiling effect (100% until noise ratio 0.8), resolved to a realistic degradation curve at n=300. The pipeline also demonstrated strong **negative rejection** — FinQA: 93.3%, TatQA: 86.7% — reliably refusing to answer when context is entirely noise rather than hallucinating from parametric memory. This is a critical safety property for enterprise deployment.

7. Conclusions

This project evaluated RAG pipelines across 134+ configurations in four domains, governed throughout by a single-variable experimental discipline. The principal conclusions, each established through independent replication:

1. **Judge independence is non-negotiable.** Self-evaluation produces systematically biased hallucination detection. This has immediate implications for any production evaluation pipeline where the generator also grades its own output.
2. **Structural engineering outperforms prompt engineering.** In the legal domain, 24 prompt-engineering configurations could not simultaneously maximise Adherence and Utilization. A single structural change — Dual-Track JSON decoupling — broke the tradeoff and achieved what brute-force prompt tuning could not.
3. **Domain transfer requires validation, not assumption.** In every domain where a configuration was transferred from another, direct testing found significant deviations from expected results. Configuration transfer assumptions fail more often than they hold.
4. **Ground-truth quality is an upstream constraint.** RAGBench's own reference responses can contain errors. Low AUCROC at small n should never be read as proof of pipeline failure without verifying the underlying reference responses.
5. **RAG evaluation is not a solved problem.** The judge's scaffolding-sentence leniency inconsistency — independently confirmed across two domains and multiple question samples — represents a genuine, unresolved limitation of the current LLM-as-judge paradigm that no configuration change addressed.

8. Future Work

DIRECTION	RATIONALE
Scale all domains to $n \geq 80$ with fresh random seeds	Current adopted results are best-case small-sample estimates. Larger n reveals systematic RMSE degradation invisible at $n=15$.
Ground-truth-aligned sentence keying	Naive regex sentence splitter creates systematic boundary mismatches. A validated fix was never ported to the production notebook — porting would reduce residual RMSE.
Sentence-window chunking for FinQA	The one retrieval-side lever never directly tested on FinQA; the legal domain showed chunk size is meaningful.
Retrieval-miss-driven hallucination detection	When the retriever fails to surface the needed document and the generator correctly refuses, the judge marks the refusal as "supported" — a failure mode invisible to any context-grounded judge.
Fine-tune generators on Dual-Track reasoning traces	The legal domain's Rich Dual-Track prompt produces structured JSON reasoning chains. Fine-tuning on these traces could embed the structural decoupling directly into model weights, eliminating the prompt overhead.
TatQA hybrid retrieval independent validation	TatQA inherited FinQA's hybrid stack without independent verification. The financial baseline showed these components may add no value over dense-only retrieval.

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